

Numerical constants:

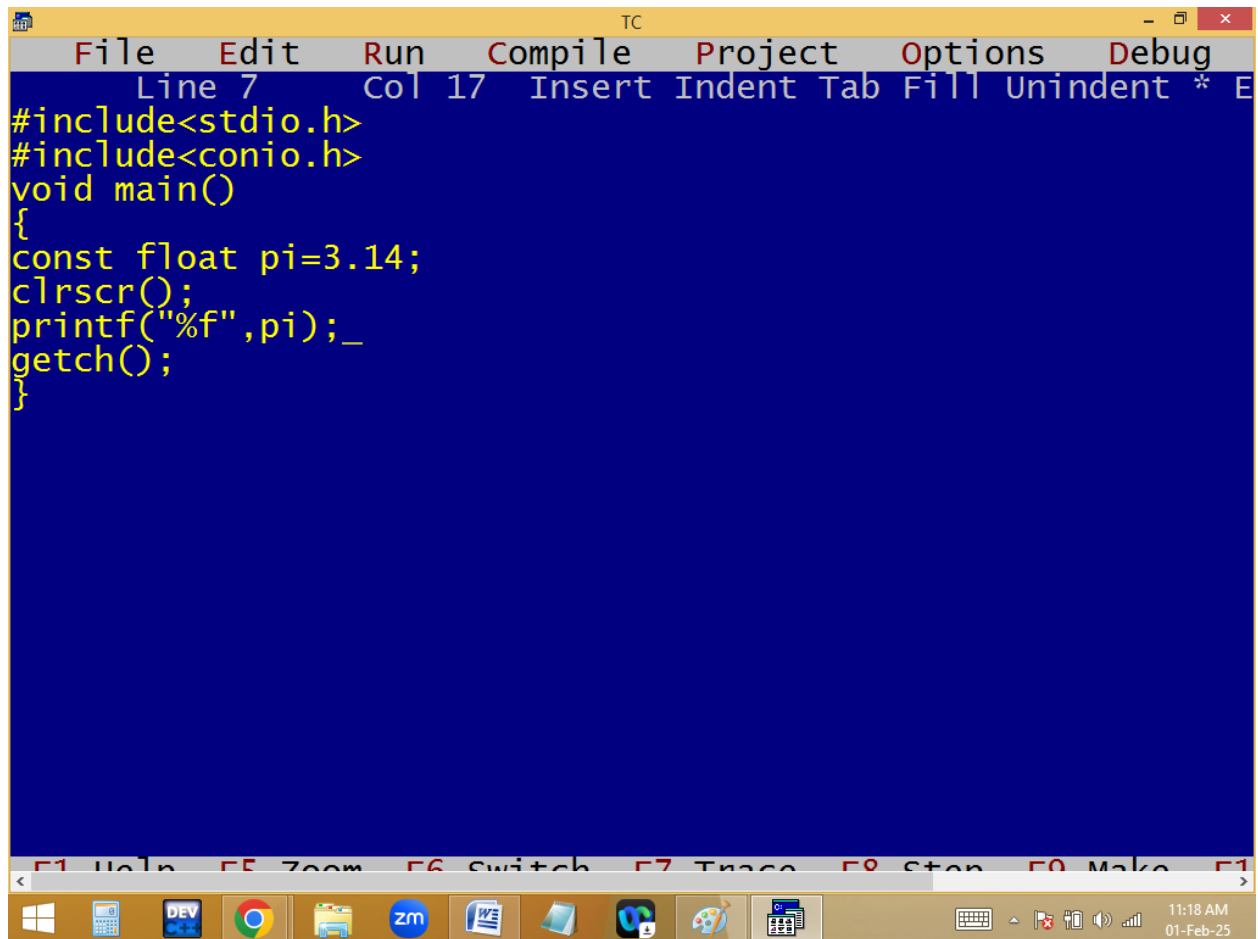
const float pi=3.14;

const int id = 1234;

Character constants:

const char name[]="Ravi"; ← string constant

const char gender = 'M'; ← char constant

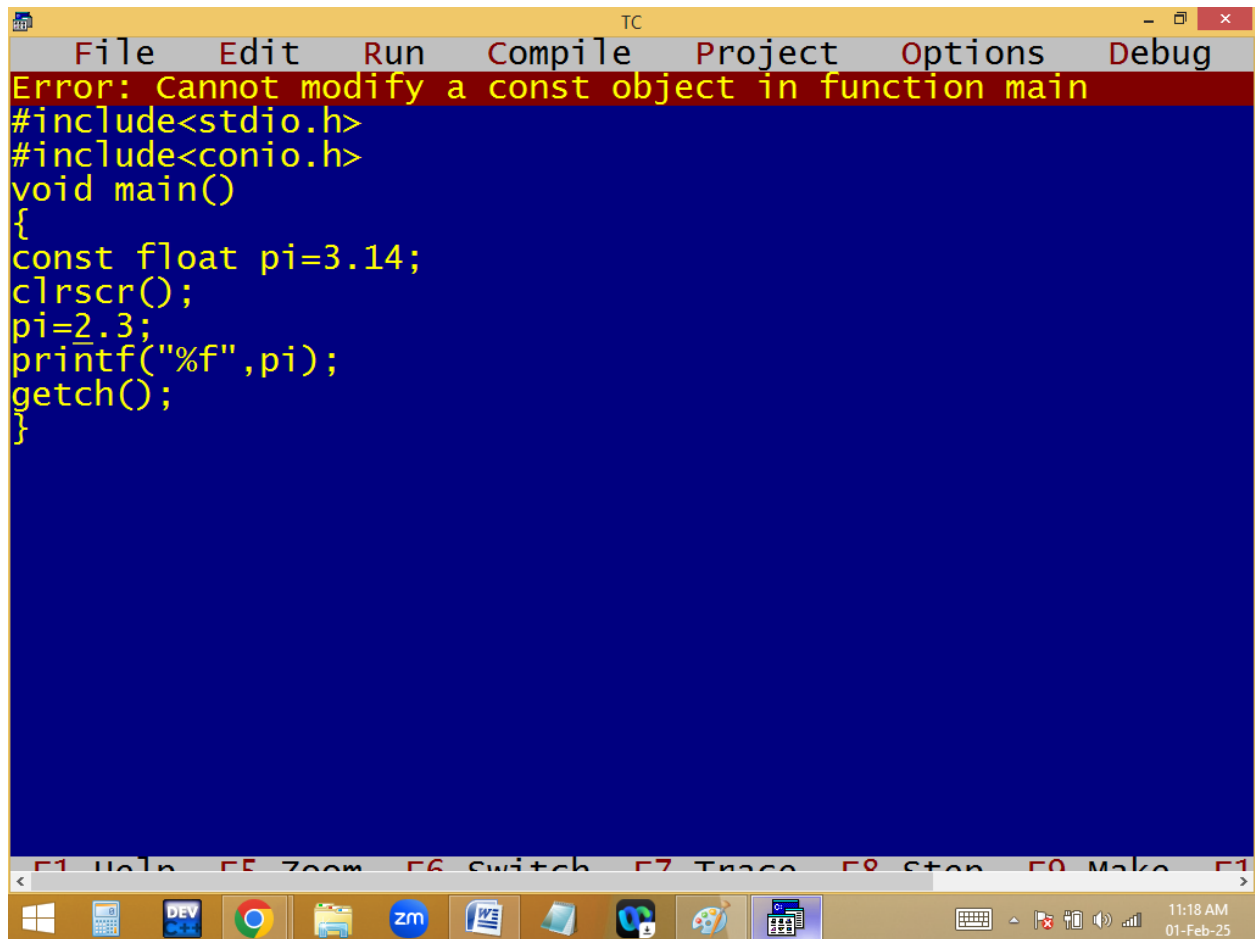


The image shows a screenshot of the Turbo C++ (TC) Integrated Development Environment (IDE). The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 7", "Col 17", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi=3.14;
clrscr();
printf("%f",pi);_
getch();
}
```

Below the code, there is a row of function key shortcuts: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock in the bottom right corner shows "11:18 AM" and "01-Feb-25".

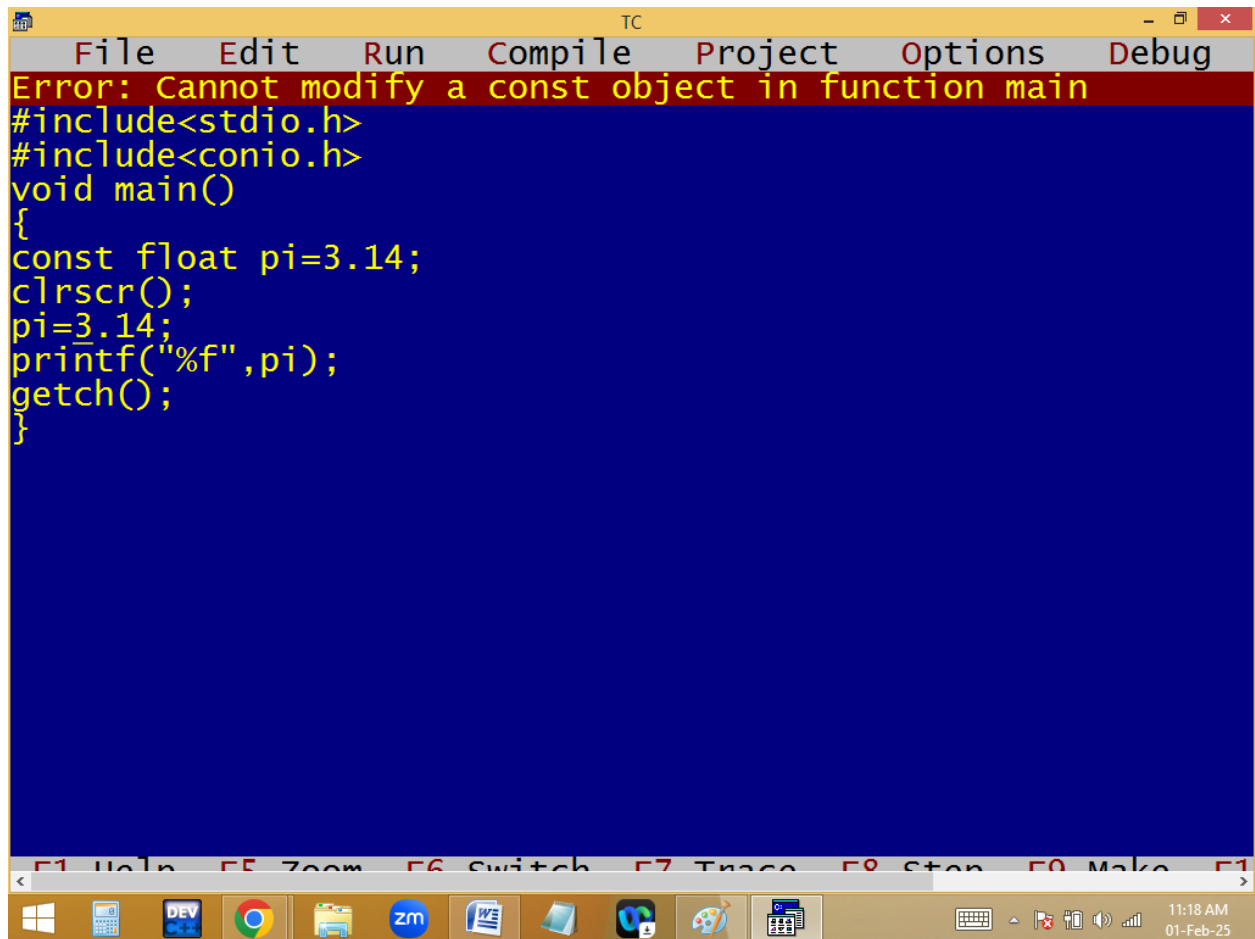




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Cannot modify a const object in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi=3.14;
clrscr();
pi=2.3;
printf("%f",pi);
getch();
}
```

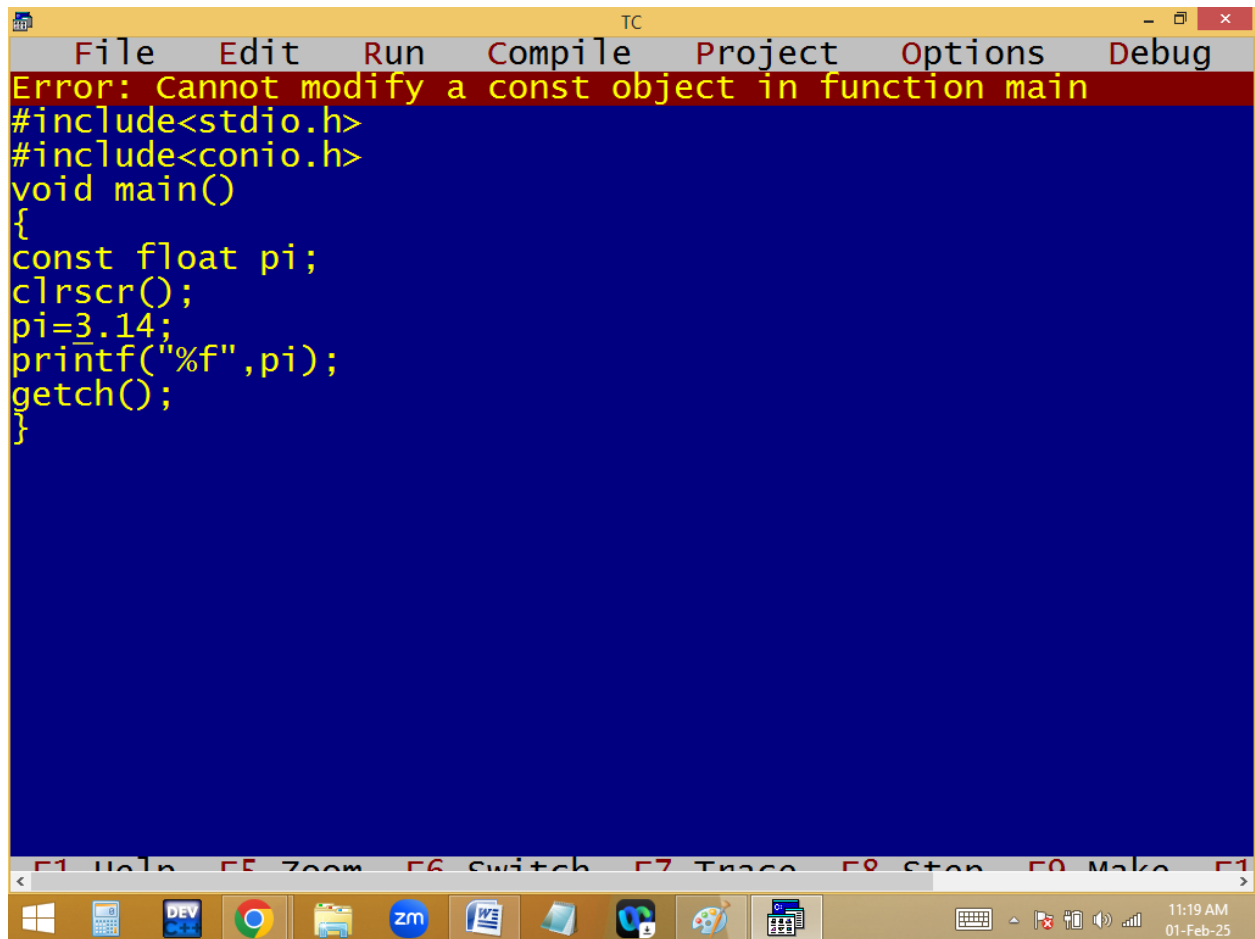
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock in the bottom right corner shows "11:18 AM" and "01-Feb-25".



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the editor area states: "Error: Cannot modify a const object in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi=3.14;
clrscr();
pi=3.14;
printf("%f",pi);
getch();
}
```

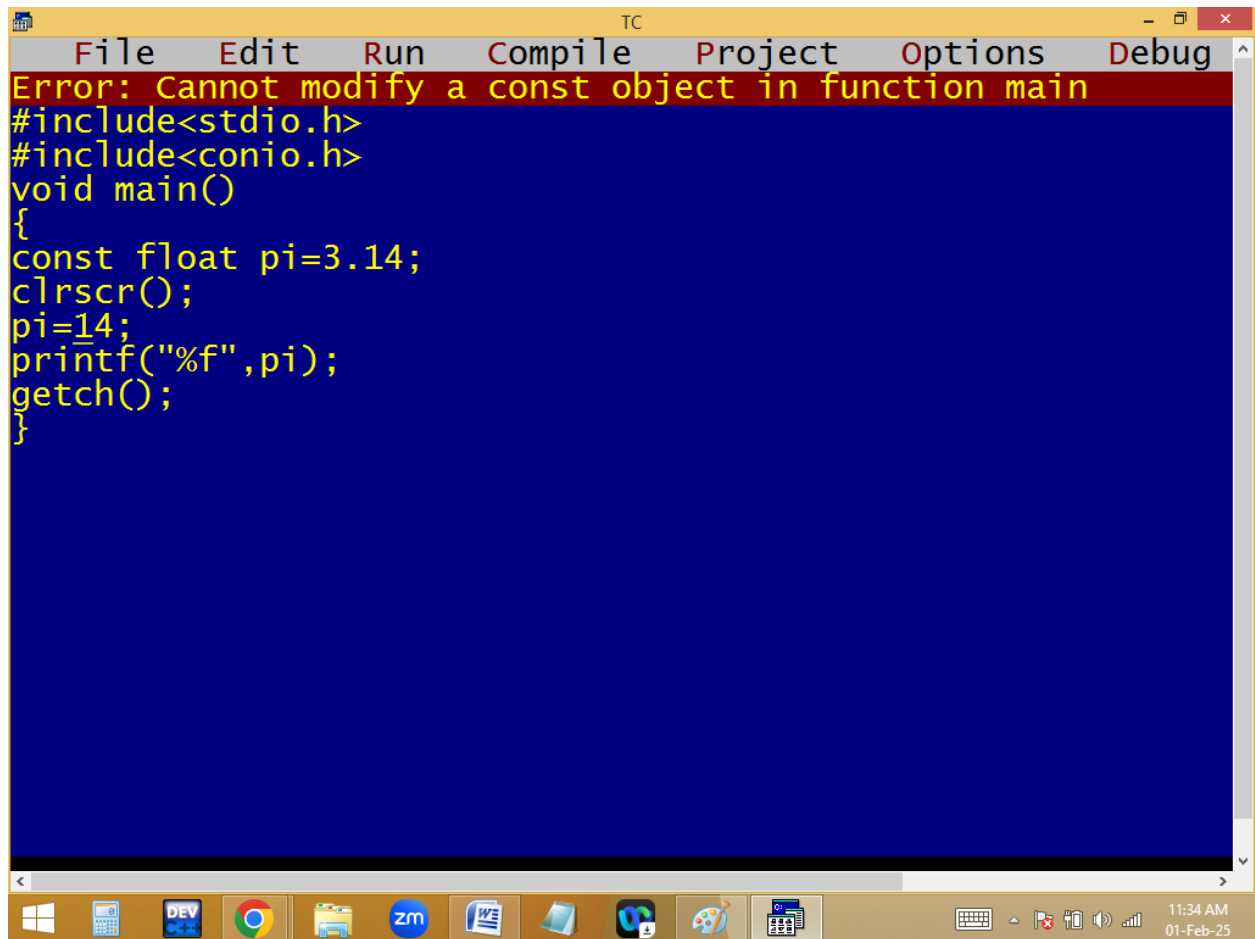
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom shows various application icons including Windows Explorer, Dev C++, Google Chrome, File Explorer, Zoom, Word, and others. The system clock in the bottom right corner displays "11:18 AM" and "01-Feb-25".



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the editor area states: "Error: Cannot modify a const object in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi;
clrscr();
pi=3.14;
printf("%f",pi);
getch();
}
```

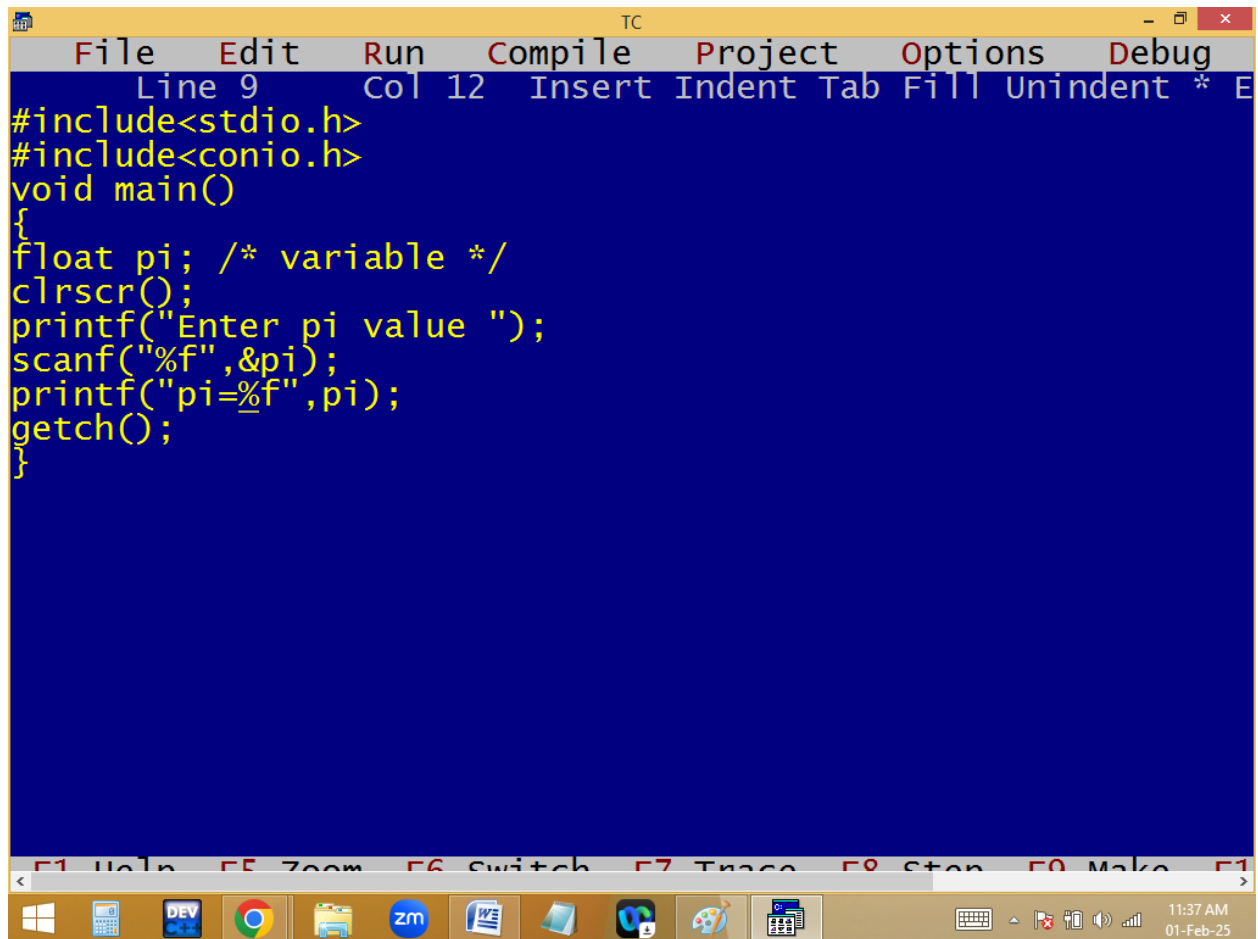
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom shows icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system clock in the bottom right corner indicates the time is 11:19 AM on 01-Feb-25.



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Cannot modify a const object in function main". Below the banner, the following C code is visible:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi=3.14;
clrscr();
pi=14;
printf("%f",pi);
getch();
}
```

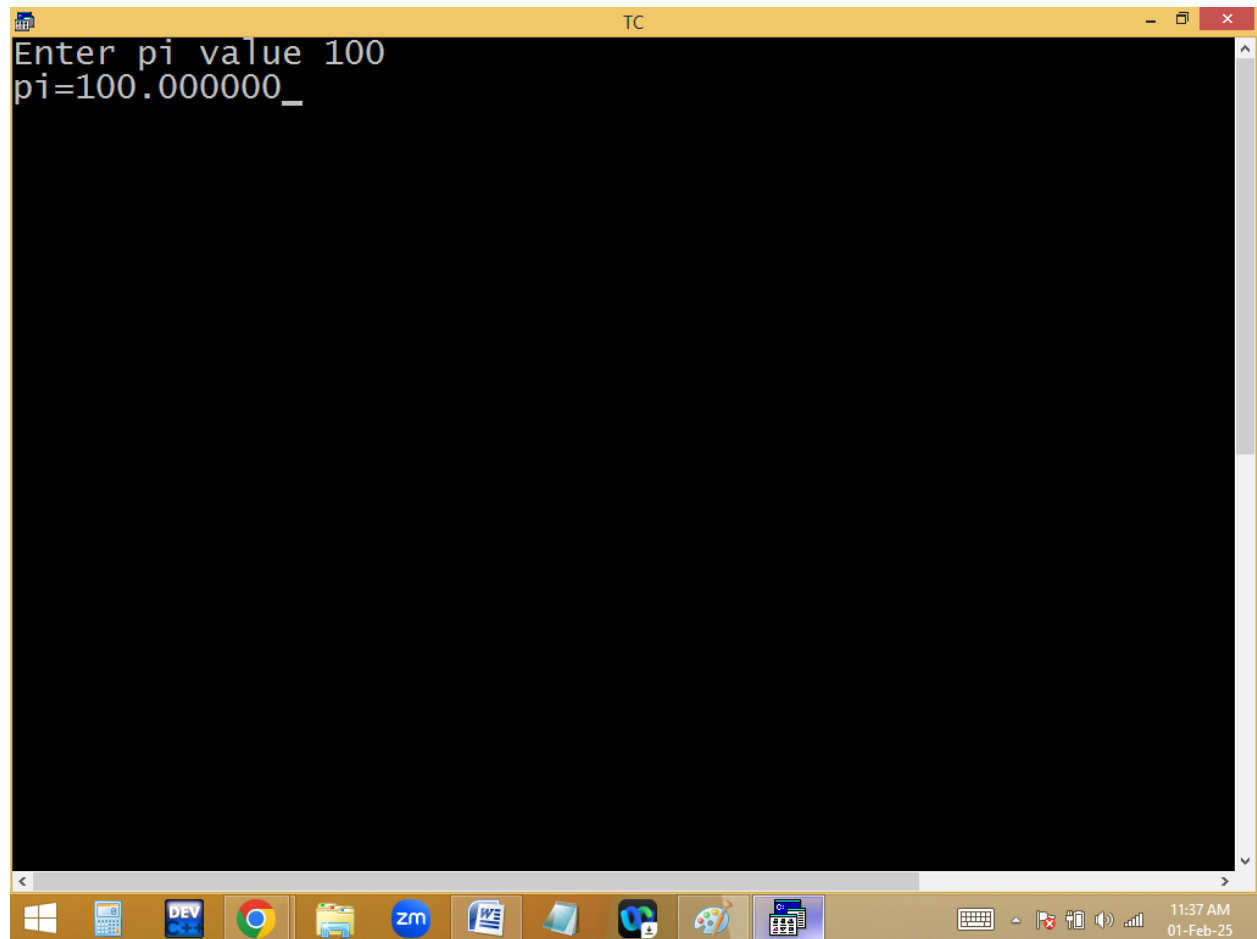
The Windows taskbar at the bottom shows the Start button and several application icons: Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, a folder icon, OneDrive, Paint, and a task view icon. The system tray on the right displays the time as 11:34 AM and the date as 01-Feb-25.

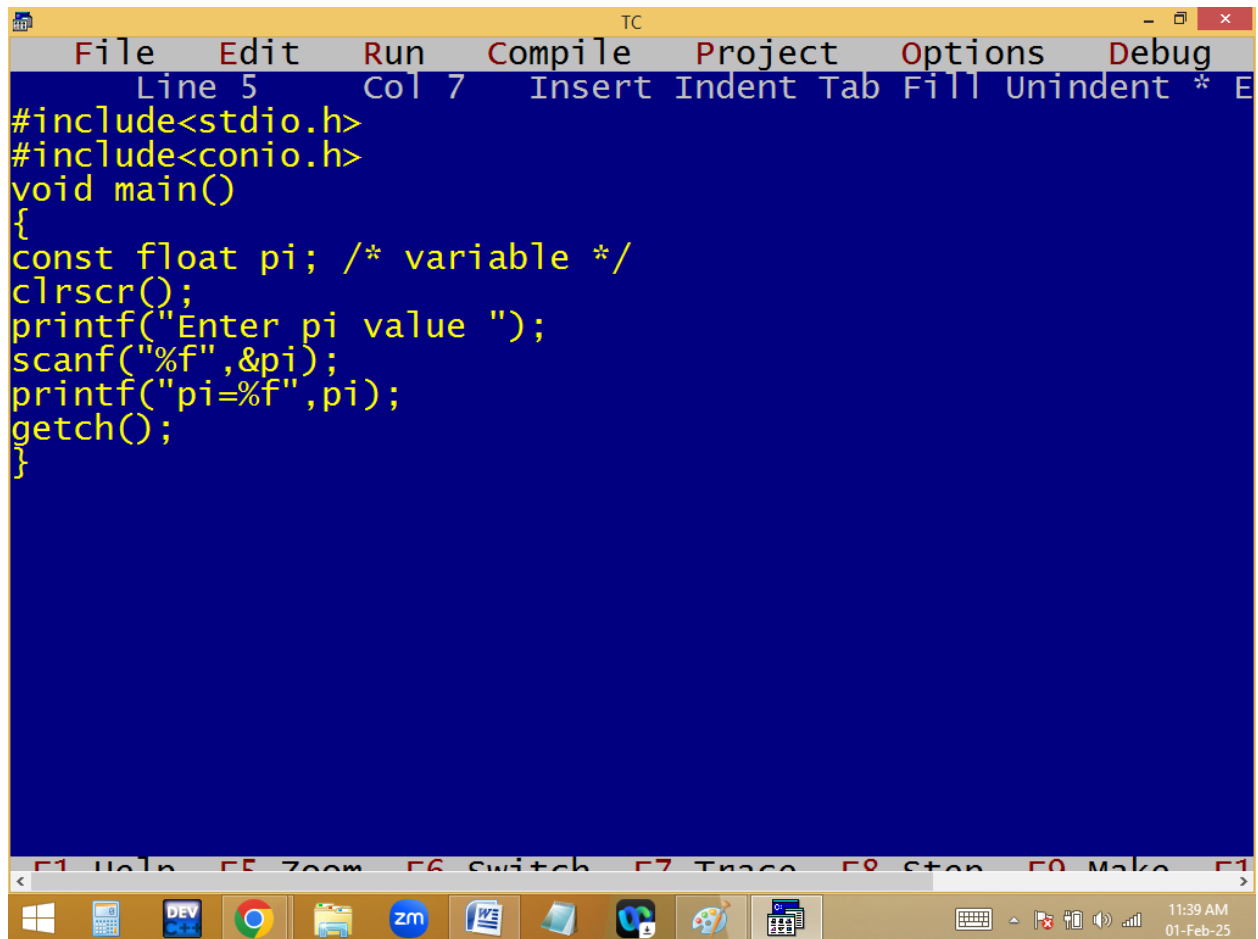


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 9" and "Col 12". The editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float pi; /* variable */
clrscr();
printf("Enter pi value ");
scanf("%f",&pi);
printf("pi=%f",pi);
getch();
}
```

Below the editor area is a toolbar with various icons and function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. At the bottom of the screen is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Dev C++, and other applications. The system clock in the bottom right corner shows "11:37 AM" and "01-Feb-25".

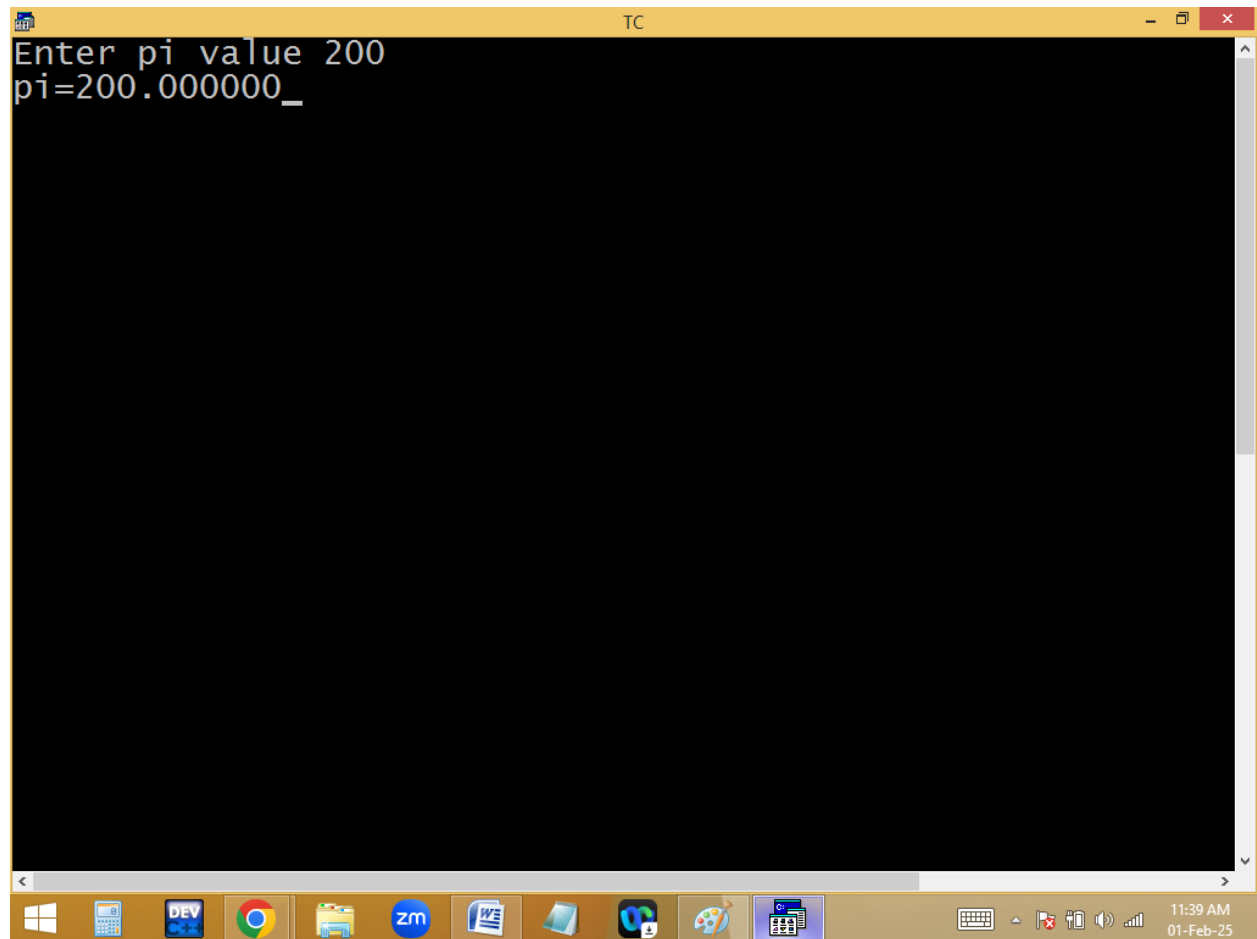


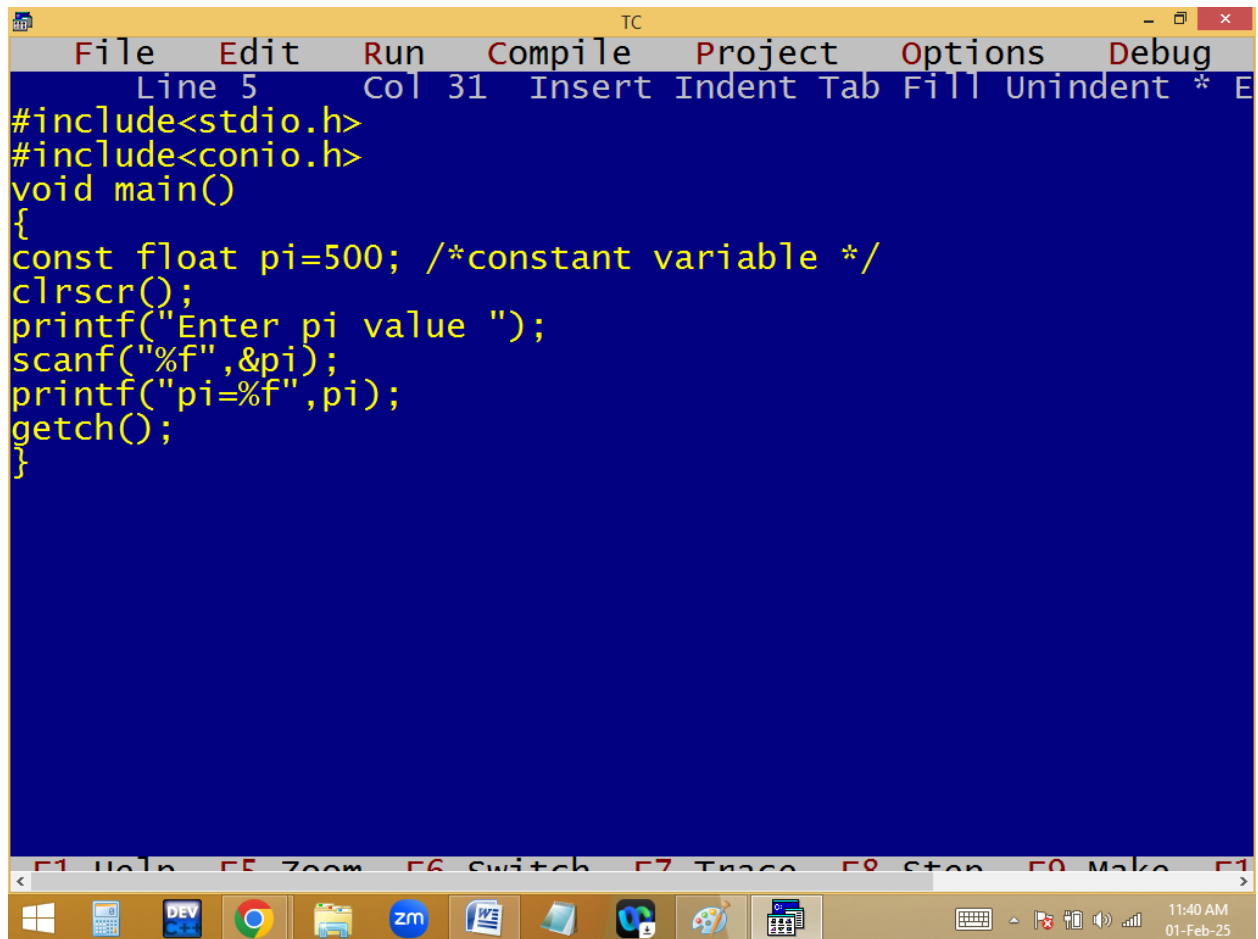


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 5", "Col 7", and a list of keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editor area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi; /* variable */
clrscr();
printf("Enter pi value ");
scanf("%f",&pi);
printf("pi=%f",pi);
getch();
}
```

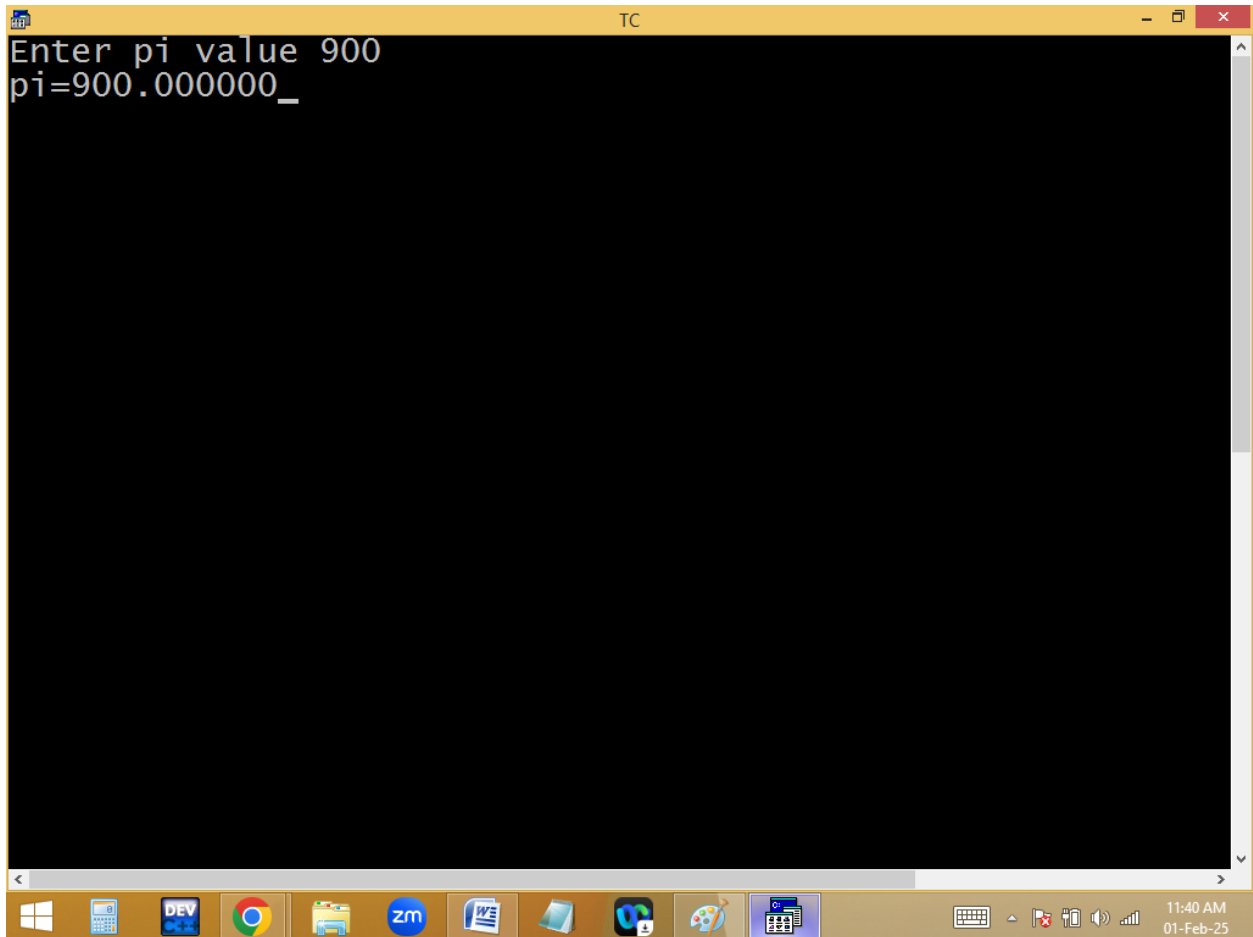
Below the editor area is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. At the bottom of the screen is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Dev C++, and other applications. The system tray on the right shows the time as 11:39 AM and the date as 01-Feb-25.





```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 31 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi=500; /*constant variable */
clrscr();
printf("Enter pi value ");
scanf("%f",&pi);
printf("pi=%f",pi);
getch();
}
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

11:40 AM
01-Feb-25



The screenshot shows a Turbo C++ (TC) window with a black background. The text "Enter pi value 900" is displayed on the first line, and "pi=900.000000_" is on the second line. The window has a yellow title bar with the text "TC". The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 11:40 AM on 01-Feb-25.

```
Enter pi value 900
pi=900.000000_
```

3. identifiers: Names of variables / functions / files / array / pointer etc.

Example for variables:

`int a=10;` ==> a used to identify 10

`int b=20;` ==> b used to identify 20

Hence a, b are the identifiers

`a=10`

`a=20`

`a=30`

a value can be changed at any time. hence a is a **variable** [change + ready]

Example for functions:

fun names

`void sum ( )`

`{`

`...;`

`}`

`void div( )`

`{`

`...;`

`}`

Example for files:

a.c, b.cpp, c.java, d.py,...

example for array:

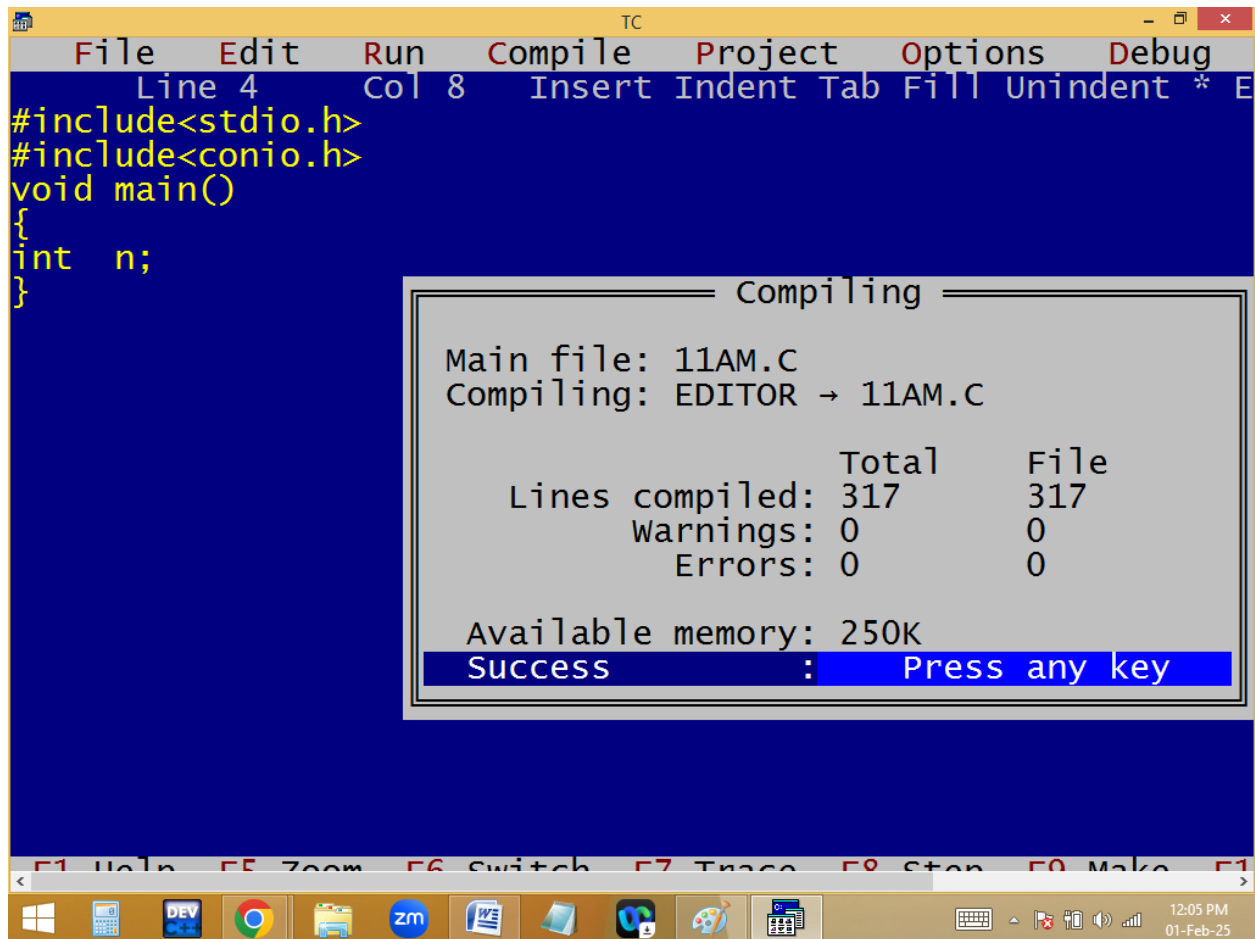
int a[10]; → int array variable

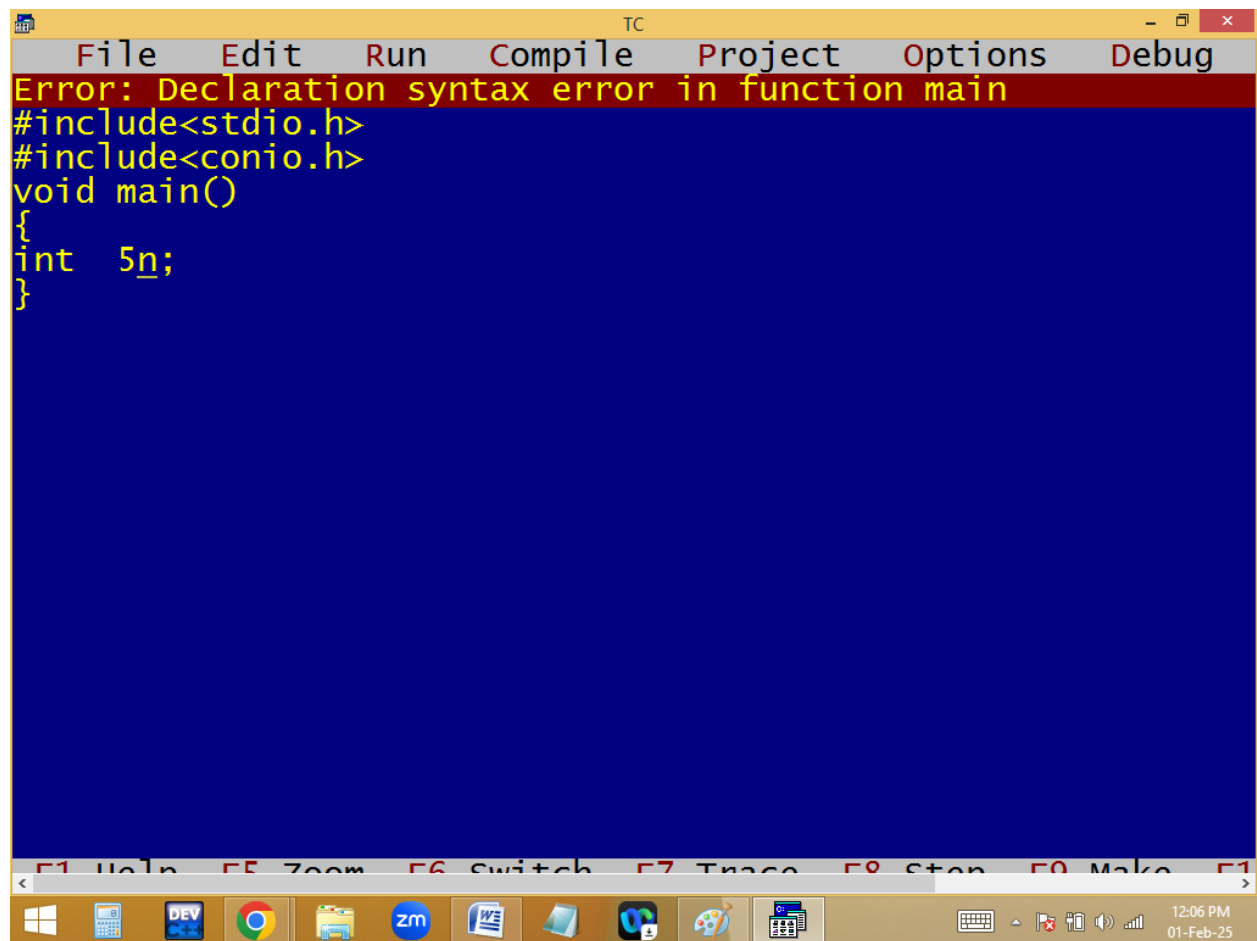
example for pointer:

int * p; → int pointer variable

identifier naming rules:

1. Identifier name should have to start with alphabet or underscore only [].





The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the editor area states: "Error: Declaration syntax error in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int 5n;
}
```

Below the editor, a toolbar contains function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom shows icons for the Start menu, Task View, File Explorer, Google Chrome, Dev C++, and other applications. The system clock in the bottom right corner displays "12:06 PM" and "01-Feb-25".

TC

File Edit Run Compile Project Options Debug

Line 5 Col 12 Insert Indent Tab Fill Unindent * E

```
#include<stdio.h>
#include<conio.h>
void main()
{
int _n ;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	317	317
Warnings:	0	0
Errors:	0	0

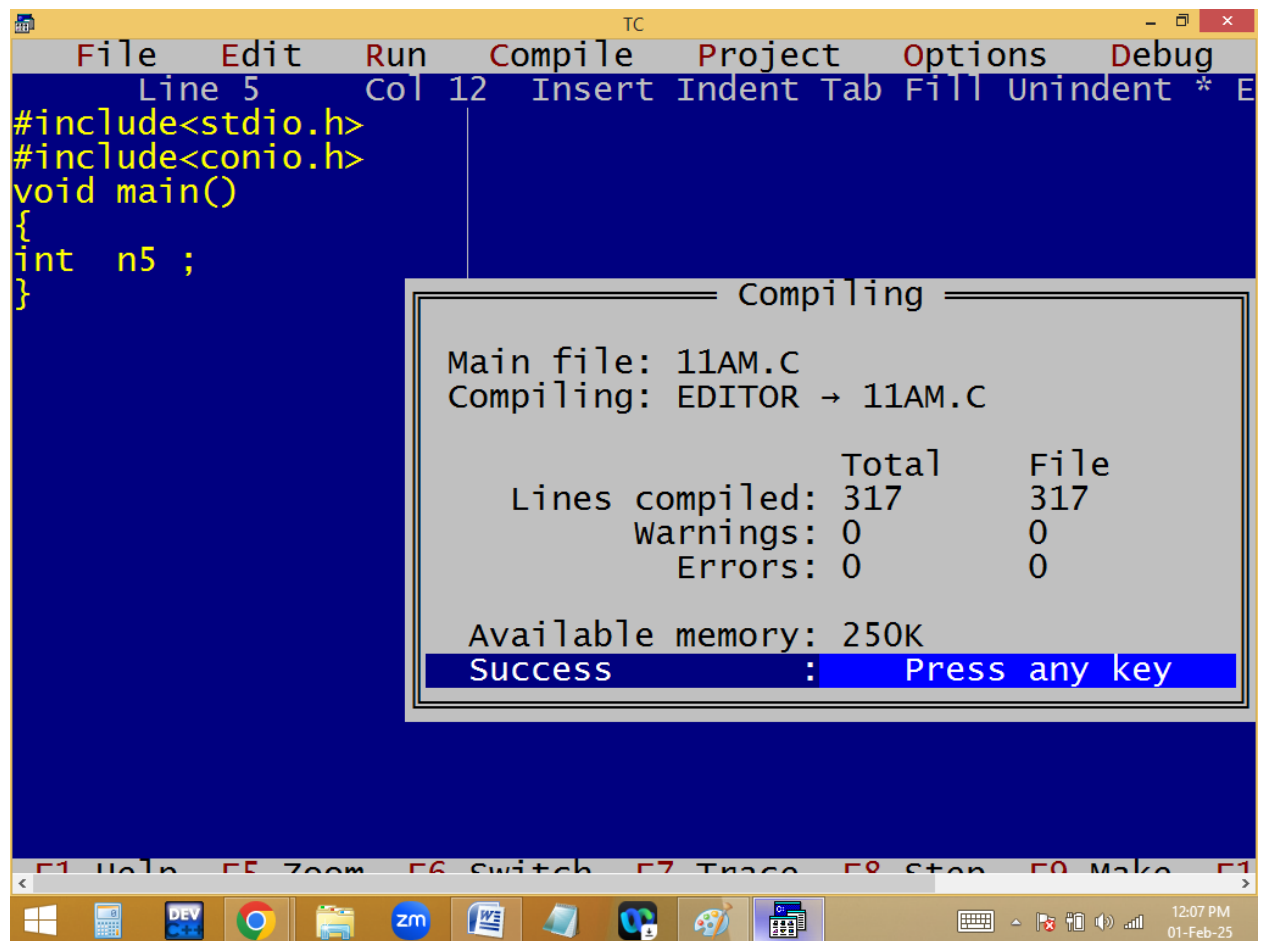
Available memory: 250K

Success : Press any key

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:06 PM 01-Feb-25

2. Numbers allowed but not at first position.



The screenshot shows the Turbo C++ (TC) IDE interface. The main editor window has a dark blue background with yellow text. The code visible is:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n5 ;
}
```

The status bar at the top indicates "Line 5" and "Col 12". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Compile" menu is open, showing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E".

A "Compiling" dialog box is displayed in the foreground. It contains the following information:

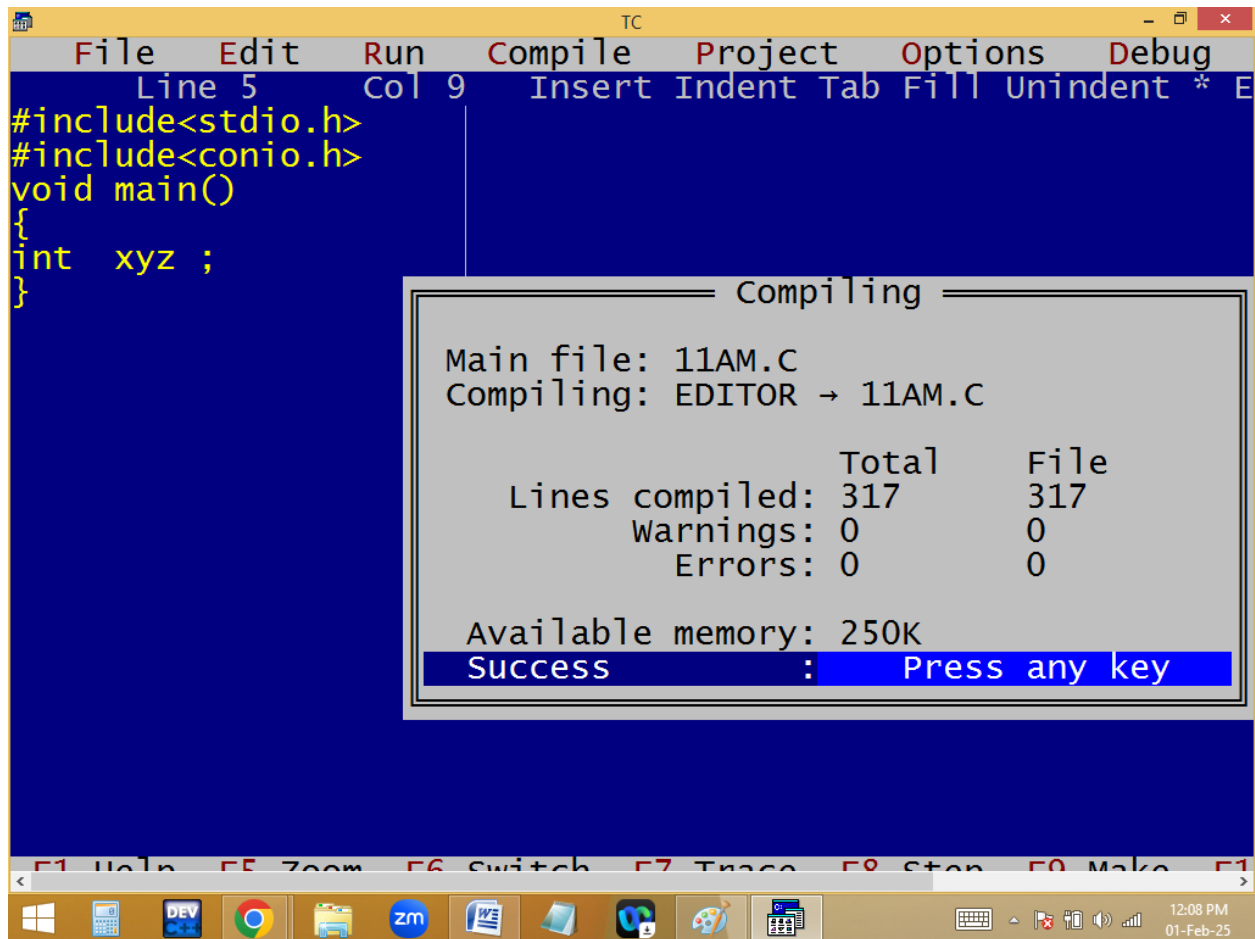
Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	317	317
Warnings:	0	0
Errors:	0	0

Available memory: 250K
Success : Press any key

The Windows taskbar at the bottom shows various icons, including the Start button, taskbar, and system tray. The system clock indicates 12:07 PM on 01-Feb-25.

3. Spaces not allowed.



The screenshot shows the Turbo C++ (TC) IDE interface. The main window has a dark blue background with yellow text. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the bottom shows 'Line 5', 'Col 10', and 'Insert Indent Tab Fill Unindent *'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  x y z ;
}
```

A 'Compiling' dialog box is open in the foreground. It displays the following information:

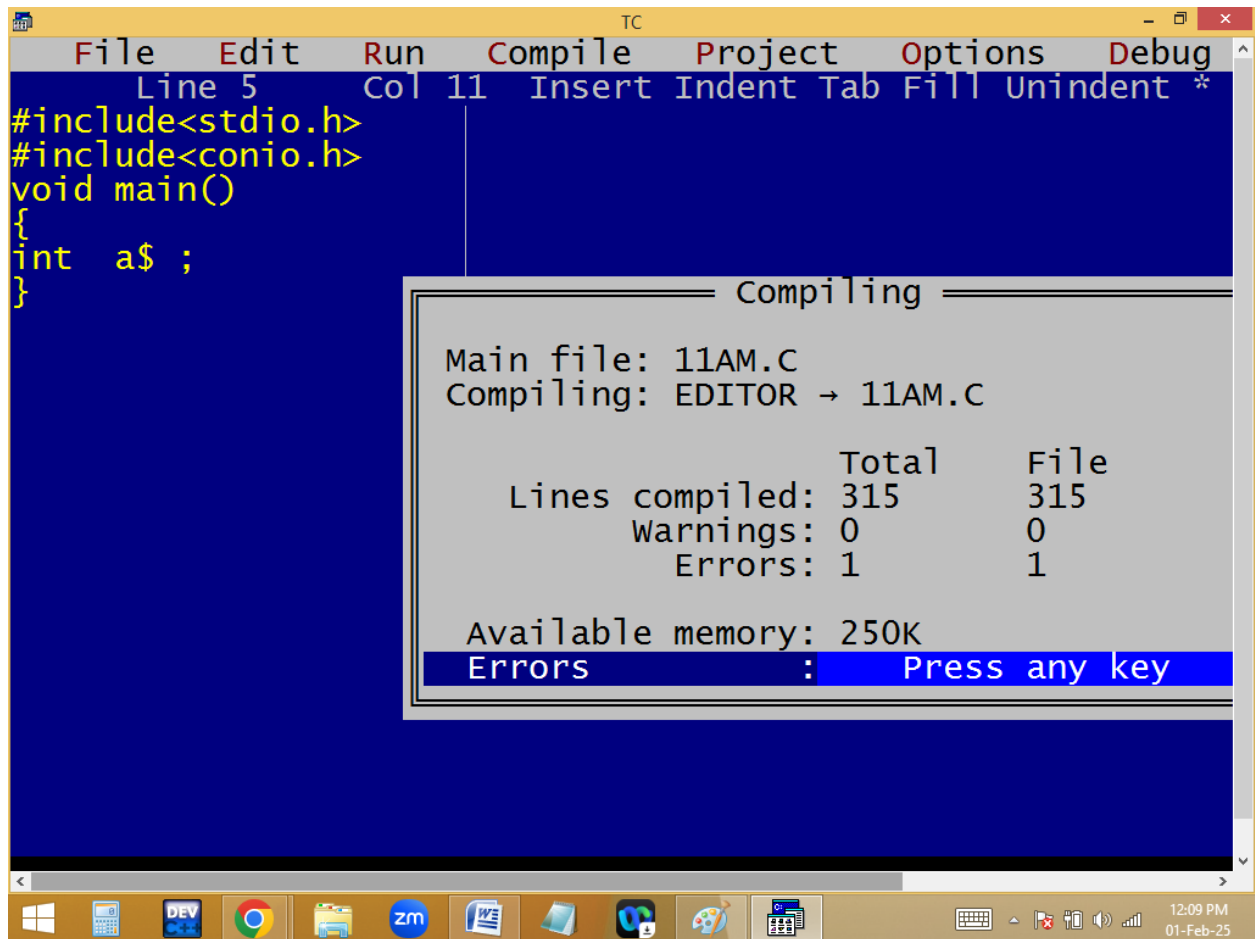
Main file: 11AM.C
Compiling: EDITOR → 11AM.C

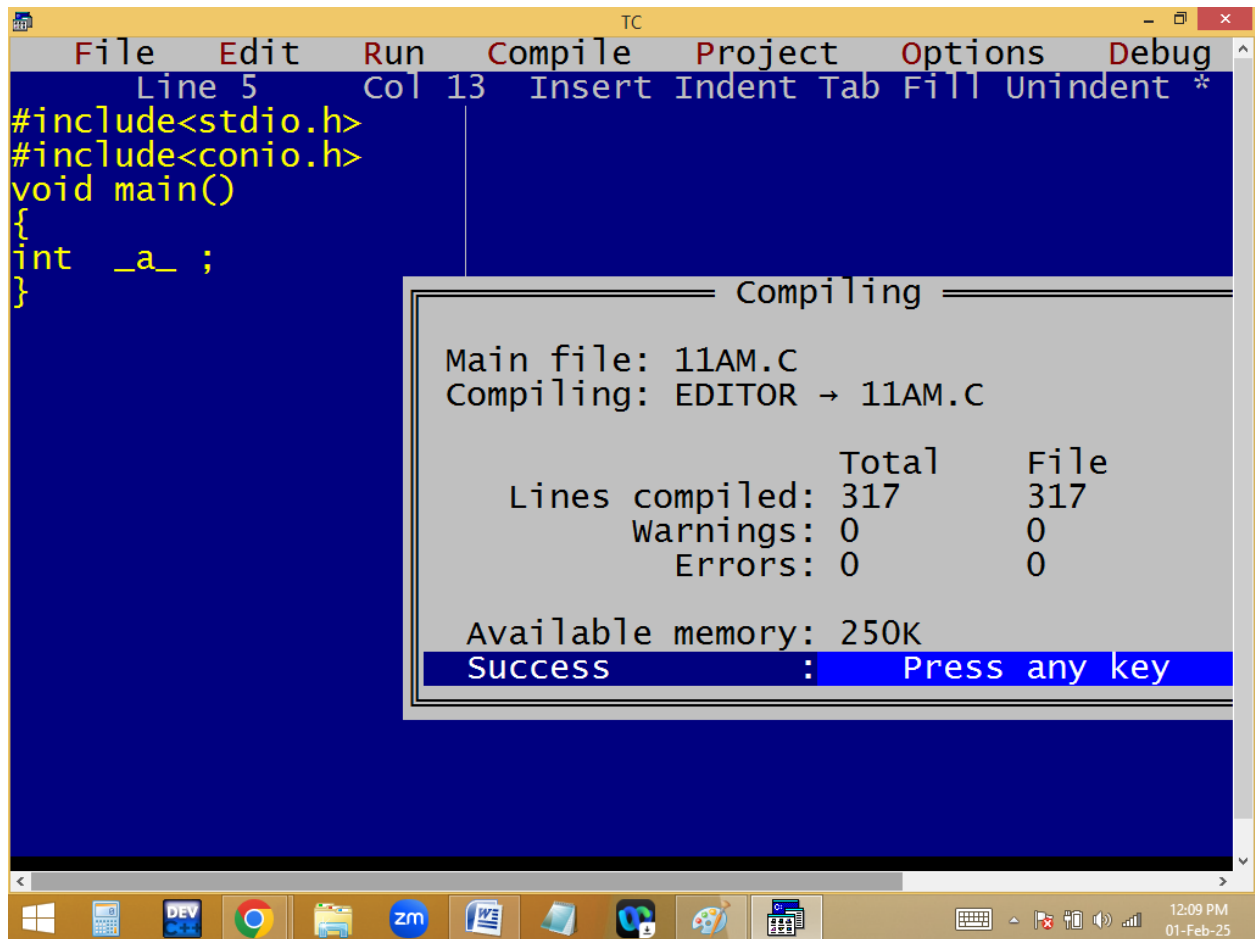
	Total	File
Lines compiled:	315	315
Warnings:	0	0
Errors:	1	1

Available memory: 250K
Errors : Press any key

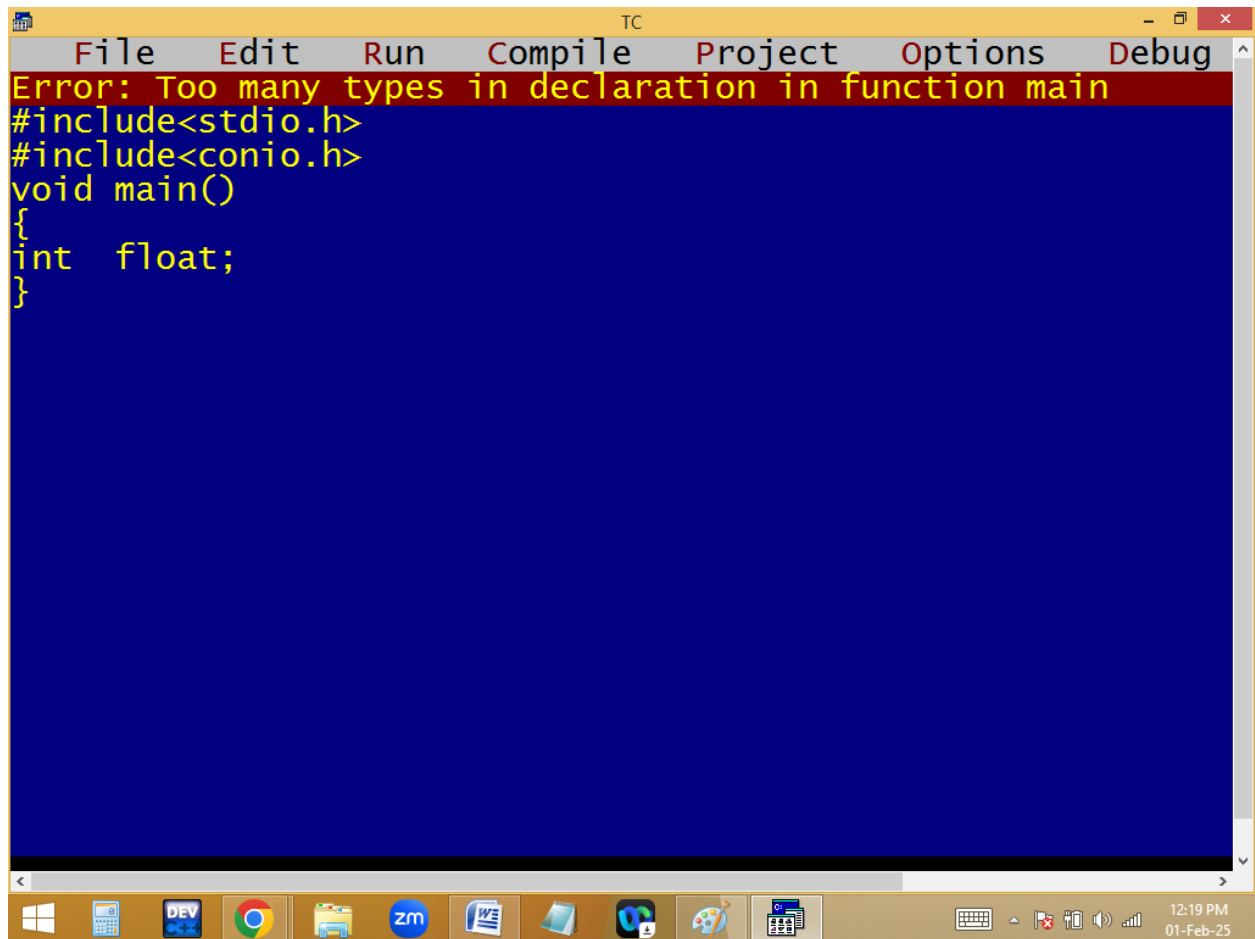
The Windows taskbar at the bottom shows various icons including the Start button, task view, and several application icons. The system clock in the bottom right corner indicates 12:09 PM on 01-Feb-25.

4. Special char's not allowed except _.





5. No keywords. i.e. name should be user defined.

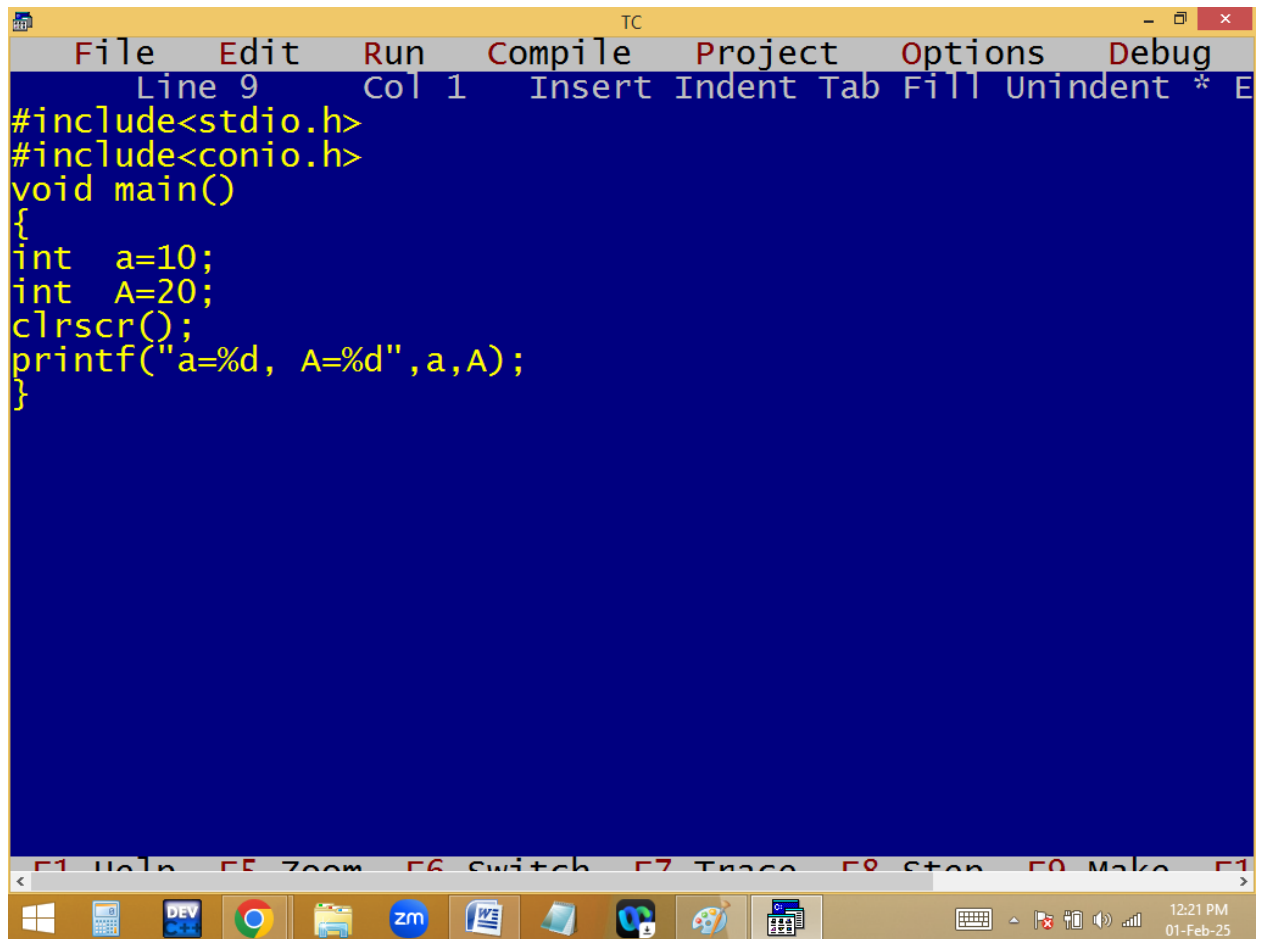


The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, and Debug. A red error message banner at the top of the code editor reads: "Error: Too many types in declaration in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int float;
}
```

The Windows taskbar at the bottom shows various application icons, including Windows Explorer, Chrome, and Zoom, along with the system clock indicating 12:19 PM on 01-Feb-25.

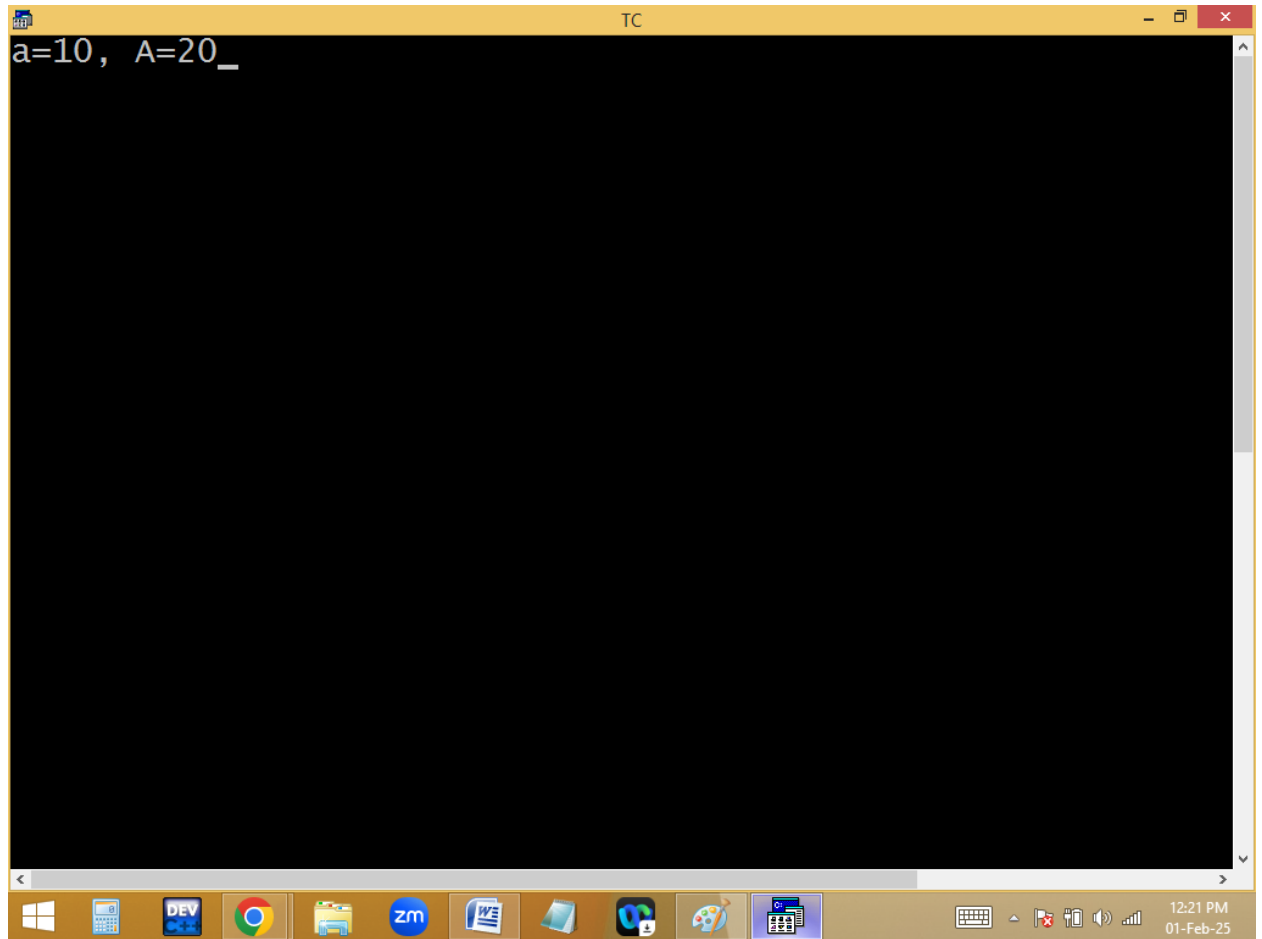
6. Identifier names are case sensitive. i.e. lower and upper are different



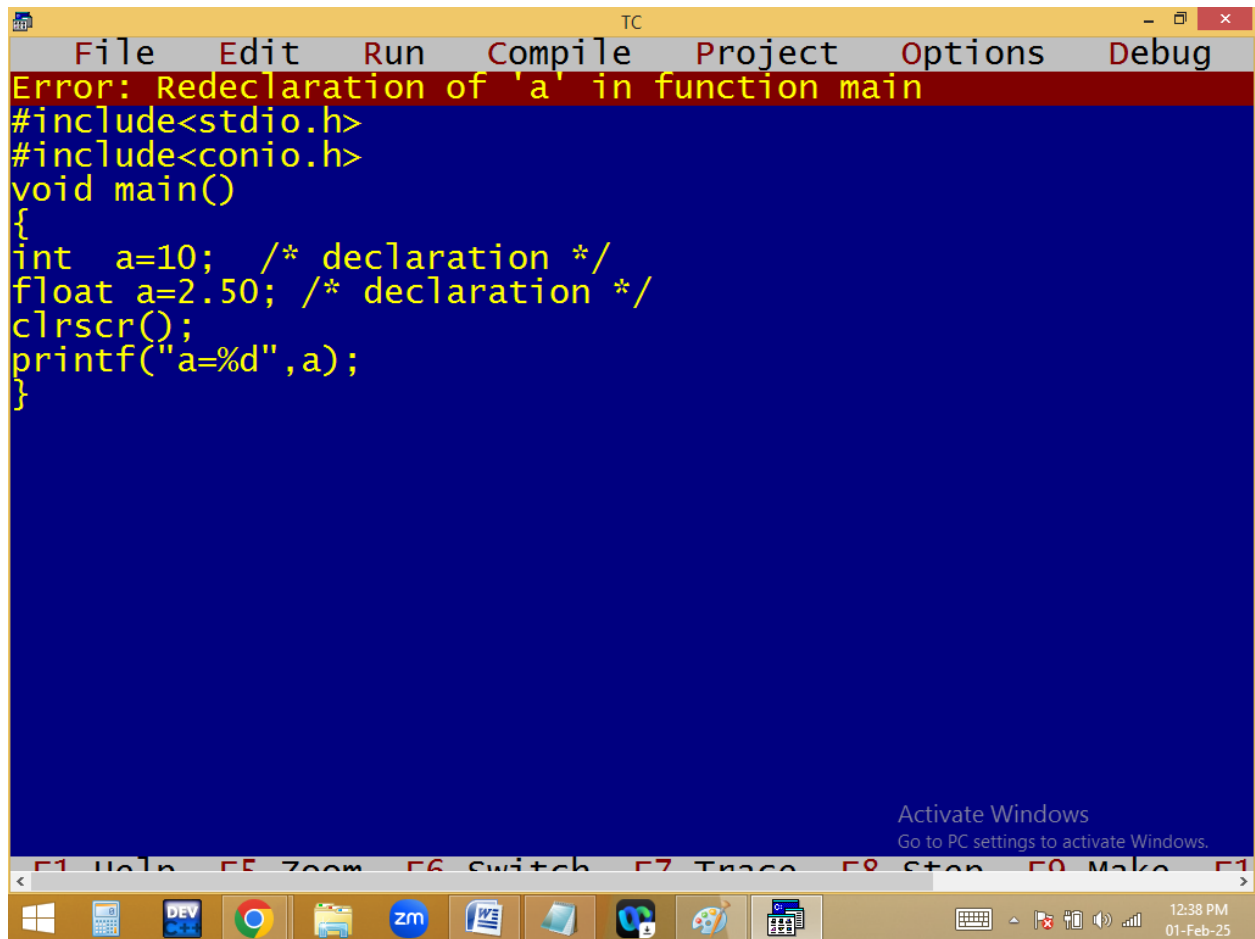
The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 9" and "Col 1". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int  a=10;
    int  A=20;
    clrscr();
    printf("a=%d, A=%d",a,A);
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "12:21 PM" and the date "01-Feb-25".



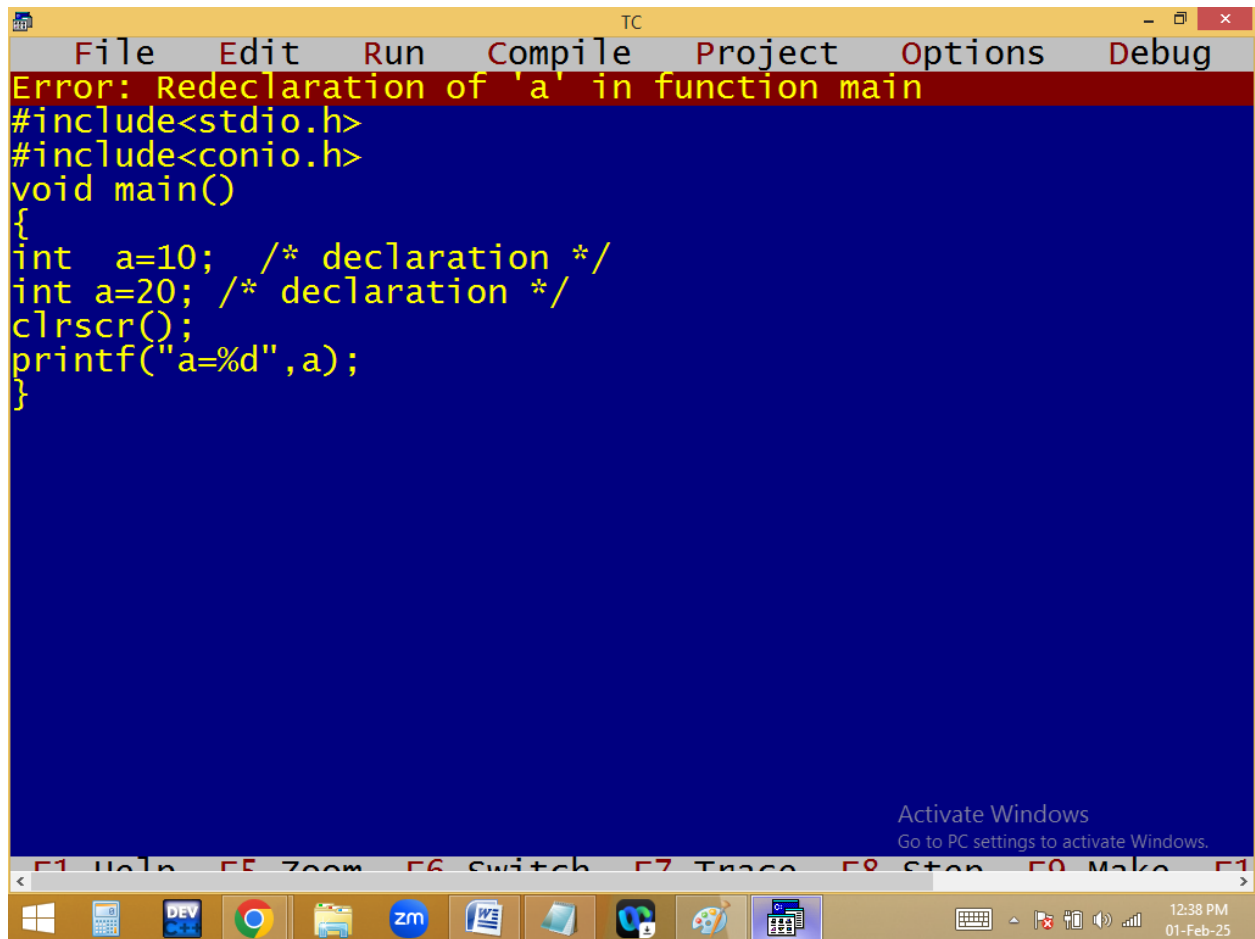
7. Duplicate identifiers are not allowed within same function or block { }.



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top reads "Error: Redclaration of 'a' in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=10; /* declaration */
    float a=2.50; /* declaration */
    clrscr();
    printf("a=%d",a);
}
```

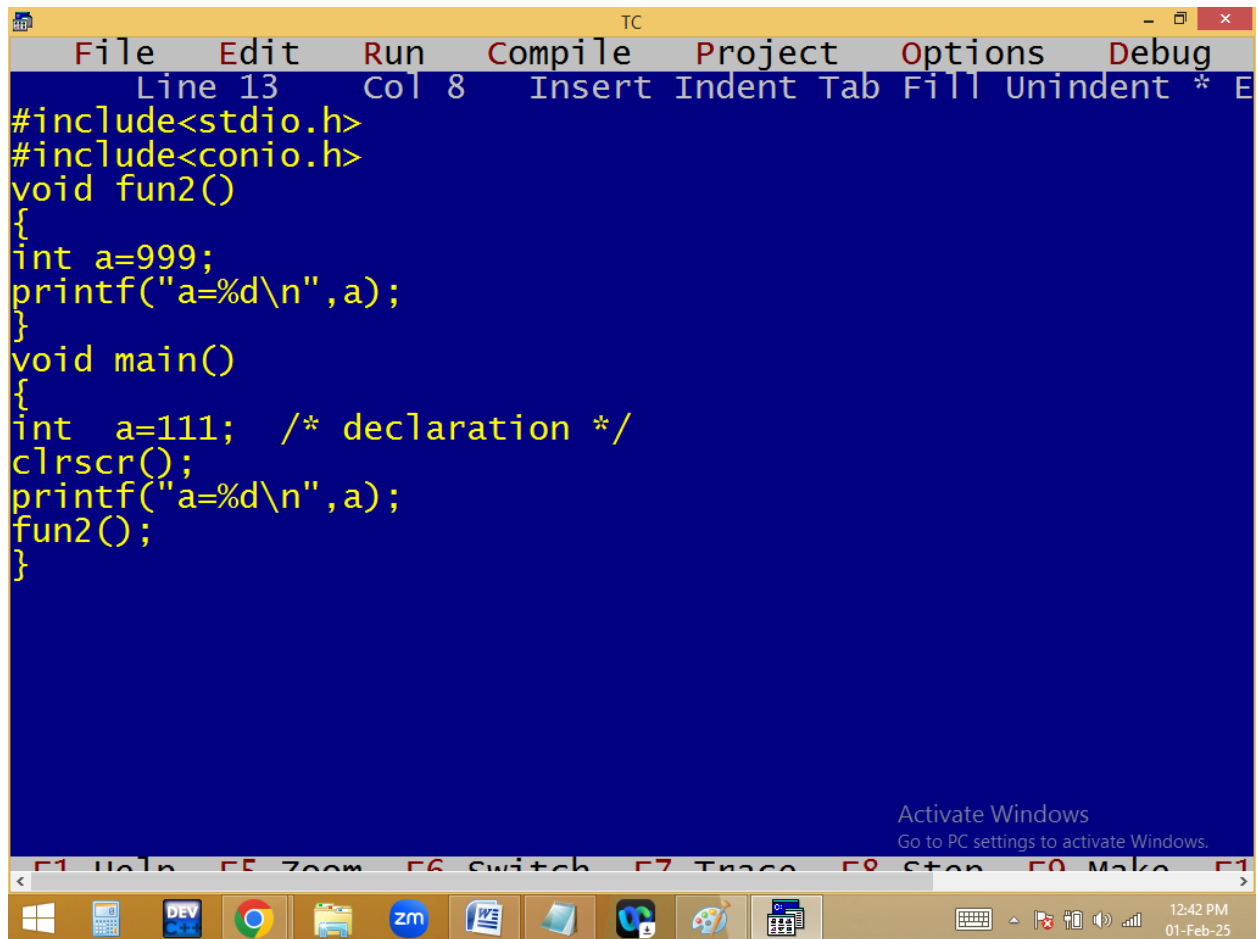
At the bottom right of the IDE, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Dev C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock shows 12:38 PM on 01-Feb-25.



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code editor displays the message: "Error: Redclaration of 'a' in function main". The code editor has a blue background and contains the following C code:

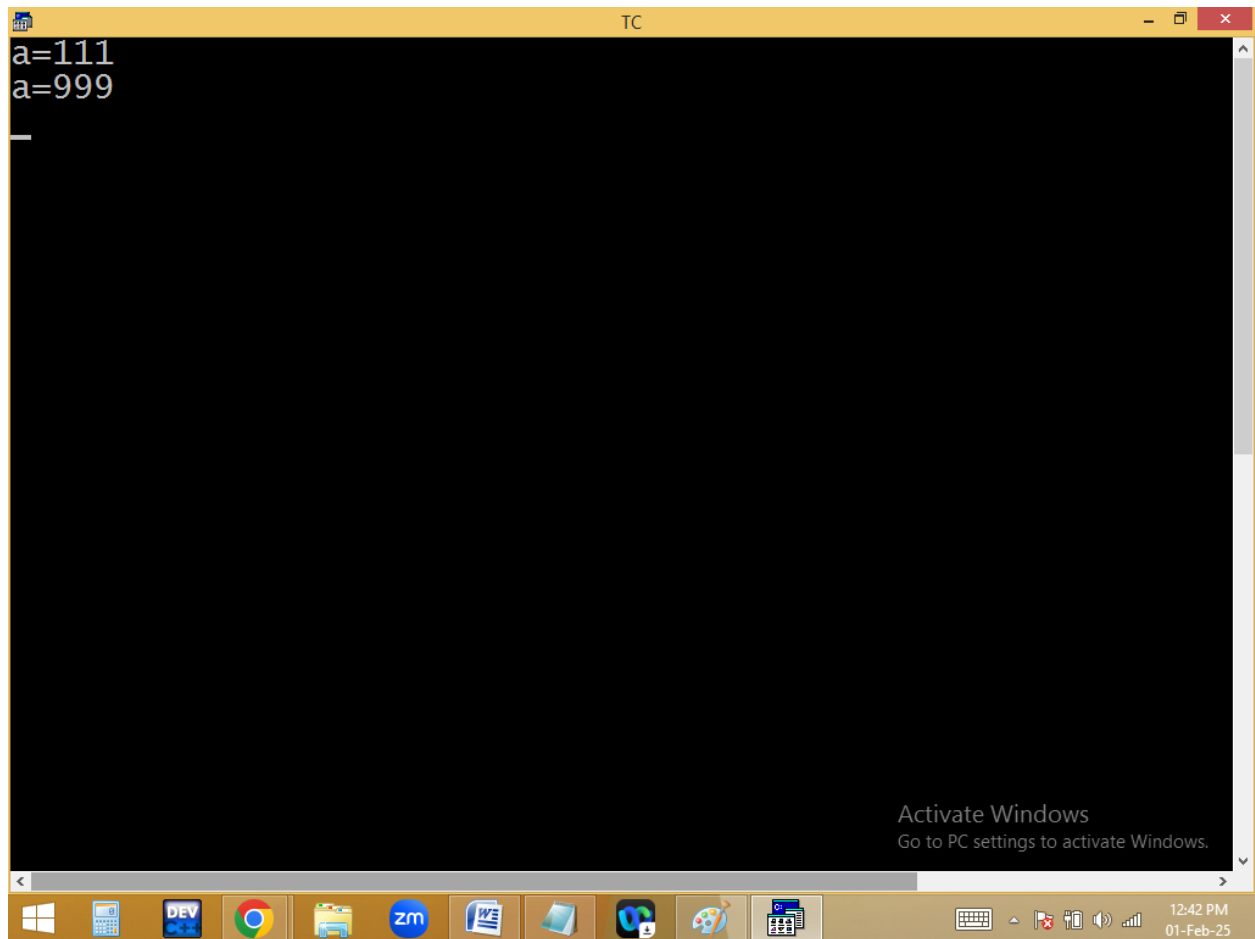
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=10; /* declaration */
    int a=20; /* declaration */
    clrscr();
    printf("a=%d",a);
}
```

At the bottom of the window, there is a taskbar with various application icons (Windows, calculator, DEV C++, Chrome, File Explorer, Zoom, Word, etc.). The system tray on the right shows the time as 12:38 PM and the date as 01-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

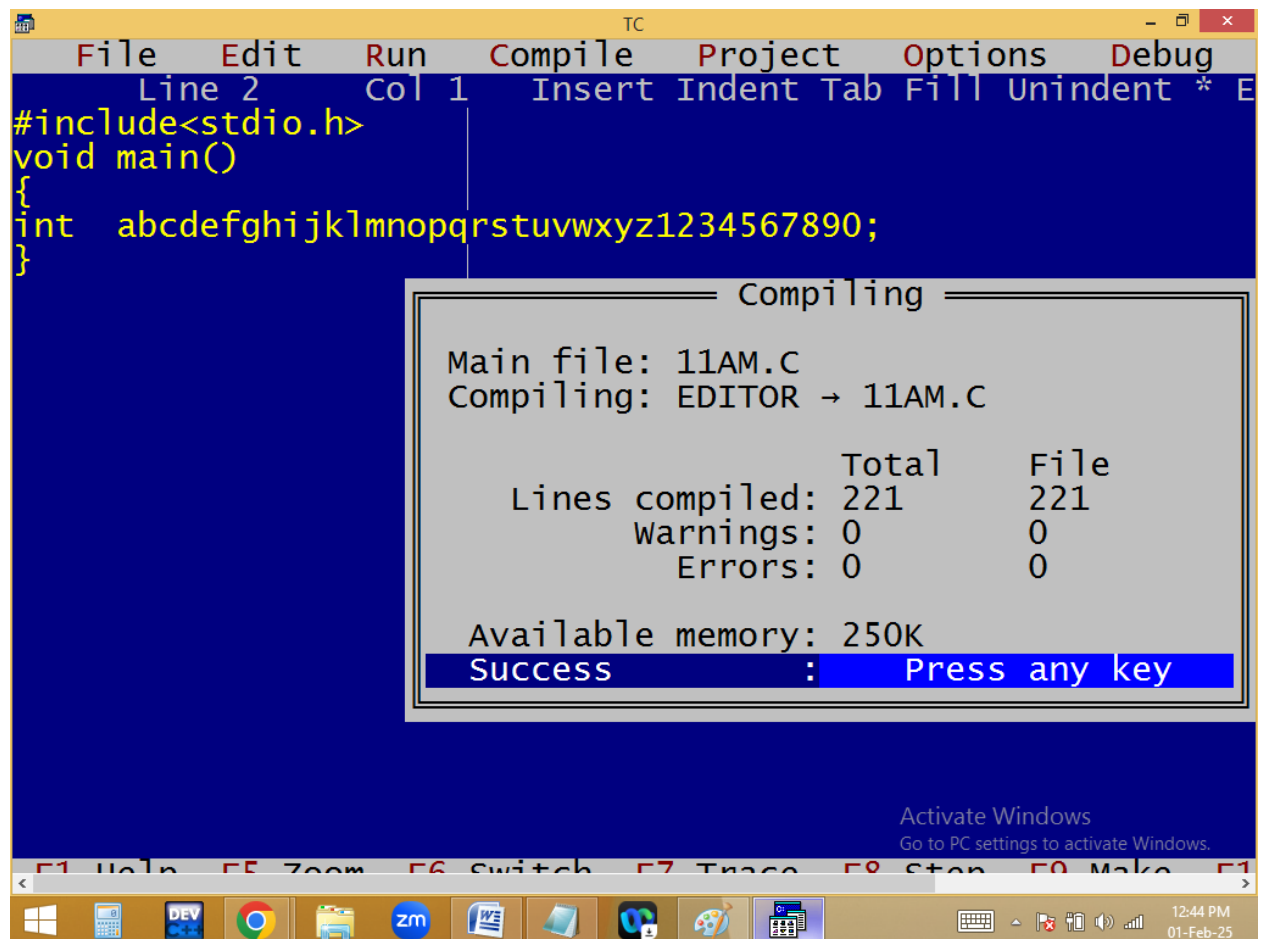


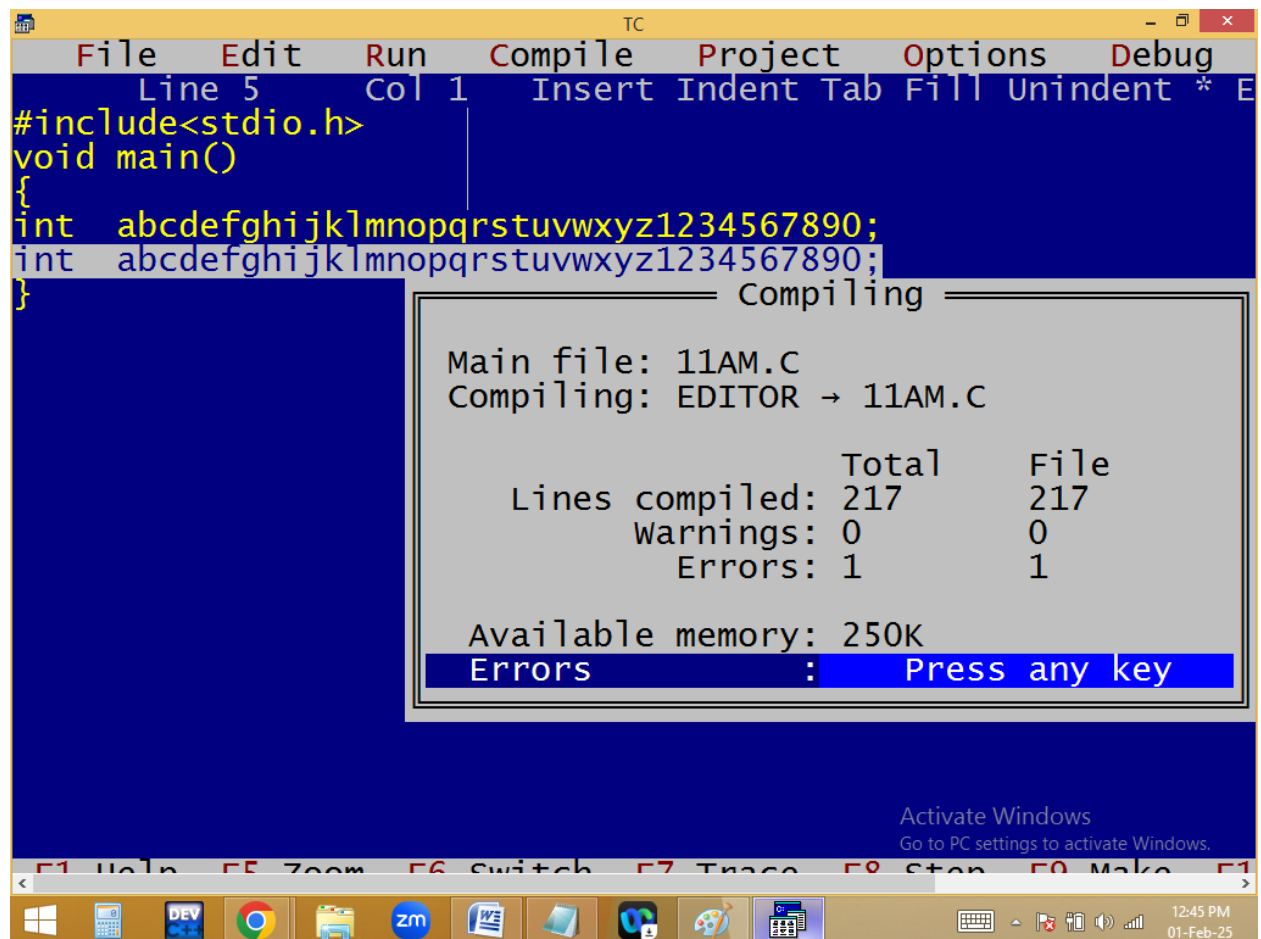
```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 8 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void fun2()
{
int a=999;
printf("a=%d\n",a);
}
void main()
{
int a=111; /* declaration */
clrscr();
printf("a=%d\n",a);
fun2();
}

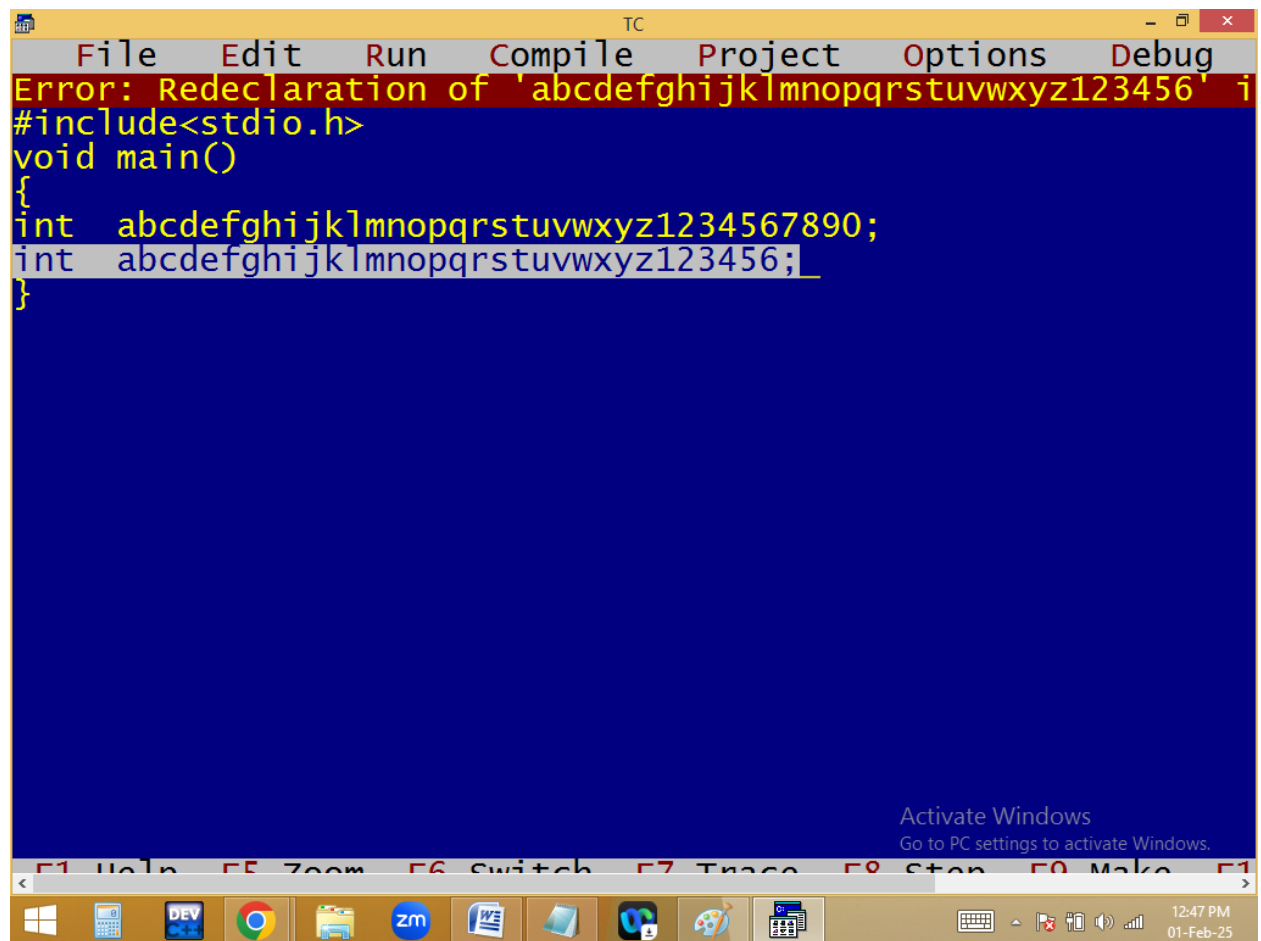
Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



8. Name may contain up to 32 char's and excess characters are ignored by the compiler.







The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The main editing area has a blue background and contains the following C code:

```
Error: Redclaration of 'abcdefghijklmnopqrstuvwxyz123456' i
#include<stdio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
int  abcdefghijklmnopqrstuvwxyz123456;_
}
```

The error message "Error: Redclaration of 'abcdefghijklmnopqrstuvwxyz123456' i" is displayed at the top of the code area. The code defines a variable with a long name in the first line of the function and then redeclares it in the second line. The Windows taskbar at the bottom shows various application icons, including Windows Explorer, Google Chrome, and Zoom, along with the system clock showing 12:47 PM on 01-Feb-25.

TC

File Edit Run Compile Project Options Debug

Line 5 Col 37 Insert Indent Tab Fill Unindent * E

```
#include<stdio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
int  abcdefghijklmnopqrstuvwxyz12345;
}
```

== Linking ==

EXE file : 11AM.EXE
Linking : LIB\CS.LIB

	Total	Link
Lines compiled:	222	PASS 2
Warnings:	0	0
Errors:	0	0

Available memory: 250K

Success : Press any key

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:47 PM
01-Feb-25

